

# 1 Introduction

Cycling has increasingly become a key component of sustainable transport policy as cities seek to improve accessibility and public health and reduce greenhouse gas emissions. In response, policymakers have introduced a range of strategies to encourage bicycle use. However, before such strategies are implemented, their potential impacts on travel behavior should be evaluated systematically. Compared to car trips, while travel time and distance remain important, bicycle route choice and mode choice are also strongly influenced by bikeability, which refers to the extent to which an urban environment or route is safe, convenient, and comfortable for cycling (1). Infrastructure quality, perceived safety, comfort, gradient, and local accessibility conditions are bikeability indicators (2, 3). However, such indicators are rarely integrated directly into activity-based and agent-based models for cycling (4). This creates a methodological gap between assessments of bicycle infrastructure and simulation-based evaluations of transport policy.

This study addresses this gap through a case study of Eskilstuna in Sweden. By linking bikeability scores with MATSim simulation, the study develops a framework for evaluating cycling policies with multiple future scenarios. The research has two main objectives: 1. to incorporate bikeability measures into an activity-based model in which cycling is represented as a main network mode; and 2. to examine how alternative planning strategies may influence bicycle demand, route choice, and mode share through changes in network accessibility.

## 2 Methods

### 2.1 Bikeability-Based Bicycle Modeling in MATSim

The bicycle simulation is implemented using the MATSim bicycle extension (5). This feature extends the standard MATSim network-routing framework by allowing bicycle trips to be treated as a network-based mode rather than as a teleported mode. Consequently, cyclists can be assigned to network links and simulated together with other modes, while bicycle-specific generalized costs can be defined at the link level (6).

Bikeability is incorporated into this modeling framework using NetAScore, an open-source toolbox for computing segment-level active-mobility indicators (7). NetAScore was selected be-

cause it is open source, uses OpenStreetMap data, and provides a standardized and reproducible workflow that can be readily applied to different cities. NetAScore extracts and prepares the street network, derives relevant link attributes, applies a bicycle-specific assessment profile, and exports the evaluated network as a spatial dataset (8). The indicators and weights used to calculate the index are based on a case study in Stockholm (9), shown in Table 1.

The bikeability index is then assigned to the corresponding directed links in the MATSim bicycle network and incorporated into the generalized cost of bicycle travel. Links with higher bikeability are assigned lower costs, and vice versa. This integration allows infrastructure quality and perceived cycling suitability to influence simulated mode and route choice, rather than relying only on travel time and distance.

## **2.2 Scenario Design**

The scenario design compares alternative cycling policy strategies by classifying all links that have bike activities into four quartiles according to bikeability. The groups represent excellent, good, moderate, and poor cycling conditions. The quartile thresholds and corresponding network lengths are reported in Table 5.

Four infrastructure-improvement scenarios are evaluated: 1. upgrading the lowest-bikeability links, 2. strengthening already high-quality corridors, 3. improving a randomly selected set of links as a comparison case, and 4. prioritizing the most frequently used links. In each scenario, improvements are represented by modifying the NetAScore-derived link values, which affect bicycle route choice, generalized travel costs, and potentially mode choice. This structure supports comparison of planning strategies and helps identify cost-effective improvements to the active-mobility network.

## **2.3 Data and Preprocessing**

The transport network is extracted from OpenStreetMap and processed to create a bicycle-compatible MATSim network. Population and travel-demand inputs are derived from the Swedish national travel survey and traffic-analysis-zone-based socio-demographic data. A population synthesizer is used to generate a synthetic population for the Eskilstuna study area (10). Scaper is an activity-chain model that constructs synthetic daily activity and travel schedules while reconciling multiple sources of

travel-behavior data (11, 12). We used Scaper to generate the agents' initial daily activity chains and travel plans. These inputs provide the baseline demand representation for evaluating how alternative cycling-network improvements may influence route choice, mode choice, and accessibility.

### **3 Expected Results**

We will build the baseline model to be calibrated to make sure that the simulated baseline bicycle mode share is consistent with available observed mode share in Eskilstuna.

For each scenario, the analysis will compare the bicycle mode share, measured as the proportion of all simulated trips made by bicycle. The result is expected to show which types of infrastructure improvements produce the largest accessibility benefits and the greatest potential increase in simulated bicycle use.

### **4 Conclusion**

This research bridges a methodological gap between bicycle-infrastructure assessment tools (NetAScore) and activity-based transport simulation (MATSim), which traditionally lacks explicit representation of bikeability. This integration enables meaningful simulations not only through travel time and distance, but also through infrastructure quality, perceived comfort, and accessibility conditions.

For Cities and regions, here illustrated with Eskilstuna municipality in Sweden, the proposed approach would support evaluation of sustainable transport and active mobility strategies. The findings reveal the relative differences between scenarios, including changes in bicycle mode share, route choice patterns and generalized bicycle travel costs. Together, these insights can help support efficient and targeted active-mobility planning that accounts for infrastructure quality and accessibility rather than relying solely on travel time and distance.

## References and Notes

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## 5 Appendix

**Table 1:** NetAScore bikeability profile parameters used to calculate segment-level bikeability in Eskilstuna

| Indicator              | Weight | Main scoring rule or mapping                                                                                                                                                                        |
|------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bicycle infrastructure | 0.127  | Bicycle way = 1; mixed way = 0.9; bicycle/bus lane = 0.75; shared lane = 0.5; undefined = 0.2; no infrastructure = 0.                                                                               |
| Road category          | 0.149  | Primary = 0; secondary = 0.2; residential = 0.8; service = 0.85; traffic-calmed = 0.9; no motorized traffic = 1; path = 0.                                                                          |
| Maximum speed          | 0.150  | Scores increase as speed decreases: $\geq 100$ km/h = 0; $\geq 80$ = 0.2; $\geq 70$ = 0.3; $\geq 60$ = 0.4; $\geq 50$ = 0.6; $\geq 30$ = 0.85; $> 0$ = 0.9; 0 = 1.                                  |
| Parking                | 0.182  | Parking present = 0; no parking = 1.                                                                                                                                                                |
| Crossings              | 0.128  | Crossing-related attributes were included using the NetAScore default bicycle-profile mapping.                                                                                                      |
| Greenness              | 0.142  | Higher greenness receives higher scores: $> 75$ = 1; $> 50$ = 0.9; $> 25$ = 0.8; $> 0$ = 0.7; 0 = 0.                                                                                                |
| Other indicators       | NULL   | Pedestrian infrastructure, designated route, maximum speed greatest, pavement, width, number of lanes, facilities, buildings, water, and noise were retained in the profile but assigned no weight. |

**beginntable[h]** Accessibility quartile classes used to define the cycling-network improvement scenarios. The quartiles have approximately equal total network length, while the number of links and observed bicycle counts differ across the network.

| Quartile | Number of Links | Length (m) | Minimum | Median | Maximum |
|----------|-----------------|------------|---------|--------|---------|
| 1        | 6,615           | 156,355.84 | 0.0859  | 0.3545 | 0.4434  |
| 2        | 7,956           | 156,351.90 | 0.4434  | 0.5046 | 0.5800  |
| 3        | 9,661           | 156,355.67 | 0.5800  | 0.7366 | 0.7570  |
| 4        | 8,346           | 156,380.67 | 0.7570  | 0.7774 | 0.7978  |